AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT101
General Mathematics

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 3/Dec. /2025

Number of Exam Pages:2 (including this cover sheet)	**Time Allowed: 90 minutes**

Instructions:

- Total Marks:**60 marks**
- This exam consists of **4 questions**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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ESSAY QUESTIONS

Each question is worth 15 marks. You should answer all the questions, giving 60 possible marks.

Q–1: [3+6+6 Marks]

a) write the answer as a + bi (complex number in standard form)

i) multiply (2-9i)(9-6i)

ii) $\frac{2+5i}{5+2i}$.

b) Solve for x: $\frac{1}{3}x + \frac{1}{4} = x - \frac{1}{6}$

Q–2: [8+7Marks] solve the following:

a) $|4x-18| \geq 2$

b) $5x - 4(6 + 2x) < 2(1 - x)$ (write the solution in interval notation).

Q–3: [6+7+2] Marks] a) solve the following equations:

i) $(x-1)^2 = 7$ (use the square root property)

ii) $2x^2+2x+5=0$ (use the quadratic formula)

b) Find the domain of f(x)= $\frac{x}{x+5}$

Q–4: [8+7Marks]

a) use completing the square to find the center and the radius of the circle

$$x^2 + y^2 + 10x - 4y + 20 = 0$$

b) Find the slope and the y-intercept of the line with equation:

4x-5y=10

End of the questions.

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